M.KUMARASAMY
COLLEGE OF ENGINEERING
NAAC Accredited Autonomous Institution
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Thalavapalayam, Karur, Tamilnadu.

THE HOSPITALITY FORECAST

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report **"THE HOSPITALITY FORECAST"** is the bonafide work of **"ROOBA B (927623BAD092), NIVETHA R T(927622BAD096), SUJITH KUMAR P (927623BAD113)"** who carried out the project work during the academic year 2024-25 under my supervision. To the best of my knowledge, the work reported herein does not form part of any other project or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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2	**Objectives**	To build a dynamic dashboard in IBM Cognos that predicts customer influx, analyses revenue per room, and provides data-driven pricing recommendations to optimize hotel profitability.
3	**Dataset Description**	It provides the foundation for forecasting demand, evaluating performance, and generating pricing insights.
4	**Tools and Technologies Used**	IBM Cognos Analytics for data visualization and predictive traffic analysis, Excel
5	**Methodology**	Analyzing hospitality data using IBM Cognos Analytics for visualization and predict customer influx, pricing optimal strategies.
6	**Implementation Steps**	Processing hospitality data in IBM Cognos Analytics for visualization and insights.
7	**Results and Findings**	The dashboard revealed seasonal peaks in customer influx, helping hotels anticipate demand and pricing strategies that maximize revenue by aligning rates with occupancy trends and customer behaviour.
8	**Conclusion**	The Hospitality Forecast dashboard enables data-driven decision-making for hotel managers through clear insights into demand, revenue, and pricing. It enhances operational efficiency and profitability by leveraging predictive analytics.
9	**Future Scope**	Future enhancements could integrate real-time booking data and external market trends for more dynamic forecasting.
10	**References**	Citing relevant hospitality analytics literature and IBM Cognos documentation for methodology validation, GitHub (if applicable)

Introduction

The hospitality industry thrives on accurate forecasting, customer satisfaction, and optimized pricing strategies. With growing competition and changing traveler behavior, data-driven decision-making is essential. The Hospitality Forecast project utilizes a rich dataset to develop a comprehensive dashboard using IBM Cognos Analytics. This dashboard enables hotels to visualize patterns in customer influx, revenue generation, and service utilization. Key metrics such as occupancy rates, booking trends, and customer satisfaction indices are analyzed to support better operational planning. Predictive models highlight upcoming booking surges and cancellations, guiding dynamic pricing decisions. Ultimately, this project empowers hotel managers to enhance profitability and guest experience through actionable insights.

Objectives

- To predict customer influx using historical and seasonal trends.
- To analyse revenue per available room (RevPAR) and other key performance indicators.
- To recommend dynamic pricing strategies based on demand and customer behavior.
- To assist hotel managers in making data-driven decisions for improved profitability.
- To visualize business insights interactively using IBM Cognos dashboards.

3. Dataset Description

The dataset contains 39 columns and several thousand rows capturing hotel operations and guest behavior. It includes attributes like booking dates, average daily rate , revenue Per room , room type, customer segment , location, booking channel, total revenue, etc..

- **Website**: Kaggle
- **Data Format**: CSV
- **Total Records**: 1001
- **Total Columns**: 40

Columns Used in the Analysis:

Column Name	Description
Date	Date allocation for booking
Location	Location of the room bookings in hotels
Hotel branch	Name of the branches where hotels are located.
Occupancy_rate	Rate about occupancy of rooms in hotels
Room type	Classification of the room
Customer segment	Types of customers are stayed in hotels
bookings	Number of bookings booked in a hotels
Cancellations	Number of bookings cancelled
Average daily rate	Average cost per room per day
Revenue per room	Amount calculated for each room
Season	Types of seasons in which the booking are booked
Early check in	Customers who have early checked in the hotels
Late check out	Customers who have check out late
Customer rating	Rating of the based on their experience
Length of the stay	Duration of the stay
Room cleanliness rating	Ratings about cleanliness
Day of the week	Week of the day rooms are booked
Promotion applied	Whether the promotion is applied(yes/ no)
Lead time days	Calculate lead time days of stay
No show	Whether the show is occured or not (yes/ no)

4. Tools and Technologies Used

- **IBM Cognos Analytics** – For creating interactive dashboards and performing business intelligence analysis.
- **Microsoft Excel / CSV** – For dataset preparation, cleaning, and initial data exploration.
- **PHP** – For developing the web interface and integrating the dashboard into a web application.
- **HTML/CSS** – For designing the frontend layout and styling of the dashboard portal.
- **JavaScript** – For dynamic content rendering and user interaction handling.
- **Apache / XAMPP** – As the local server environment to host and test the application.
- **MySQL** – For managing user sessions and backend login functionalities (if applicable).

Microsoft Excel

Purpose: Used for initial data exploration, formatting, or cleaning . Excel was helpful in identifying missing values, exploring column statistics, and performing quick aggregations.

5. Methodology

- ### Data Collection

  Gathered historical hotel data including bookings, cancellations, revenue, and customer feedback from various sources to ensure a comprehensive dataset for analysis.

- ### Data Cleaning and Preparation

  Processed the raw data to handle missing values, remove duplicates, and format attributes correctly for accurate analysis and visualization.

- ### Data Analysis

  Explored the dataset to identify trends and key metrics like occupancy rate, average

daily rate, and customer segments to understand hotel performance.

- **Dashboard Development**

  Used IBM Cognos Analytics to create interactive dashboards, enabling visualization of customer influx, revenue per room, and pricing strategies.

- **Integration and Deployment**

  Embedded the IBM Cognos dashboard within a web application using PHP, allowing easy access and user-friendly interaction.

- **Validation and Testing**

  Tested the dashboard functionality and data accuracy to ensure reliable insights and smooth user experience for hotel managers and stakeholders.

Challenges in Hospitality Forecast

- Data quality and completeness issues
- Seasonal and external factors affecting demand
- Dynamic and competitive market conditions
- Variability in customer behaviour and preferences
- Integration of multiple heterogeneous data sources
- Difficulty in maintaining real-time data updates

Impact of Hospitality Forecast

- Enhanced decision-making for hotel management
- Improved pricing strategies leading to increased revenue
- Optimized resource allocation based on demand prediction
- Better customer satisfaction through personalized services
- Reduced cancellations and no-shows via proactive strategies
- Stronger competitive advantage with data-driven insights

6. Implementation Steps

7. Results and Findings

- The Hospitality Forecast dashboard effectively visualized patterns in customer influx, revealing that bookings peaked during festive seasons and weekends, helping hotels prepare resources accordingly.

- Analysis of revenue metrics like ADR and RevPAR showed strong correlations with promotions and booking channels, enabling smarter revenue optimization strategies.

- Customer segmentation and loyalty analysis revealed that returning guests and loyalty members contributed significantly to consistent revenue.

- Dynamic pricing recommendations based on demand trends led to improved occupancy rates and better profitability across multiple hotel branches.

- Feedback and rating analysis highlighted that cleanliness and service responsiveness had the greatest impact on guest satisfaction.

8. Conclusion

The Hospitality Forecast project successfully demonstrated how data-driven insights can optimize hotel operations, enhance revenue strategies, and improve guest satisfaction. By leveraging BI tools and predictive analytics, hotel managers can make informed decisions that align with market demand and customer preferences.

9. Future Scope

- **Integration of Real-Time Data:** Incorporating live data streams (e.g., online bookings, social media trends) to enable real-time forecasting and dynamic decision-making.

- **AI-Powered Personalization:** Implementing machine learning to personalize offers, predict customer preferences, and automate pricing strategies for increased guest satisfaction and profitability.

- **Mobile App Expansion:** Developing a mobile interface for hotel managers to access dashboards and insights on-the-go.

- **Global Scalability:** Extending the dashboard framework to support multi-location hotel chains with region-specific analytics.

- **Voice & Chatbot Integration:** Using conversational interfaces for querying dashboards or reports using natural language.

10.References

- **Dataset Source – Kaggle: Hospitality Forecast Dataset**

  https://www.kaggle.com/datasets/datadives/hotel-booking-analysis

- **GitHub Link**
  https://github.com/Rooba8925/projects/tree/main/hospitality%20forecast

SCREEN SHOTS

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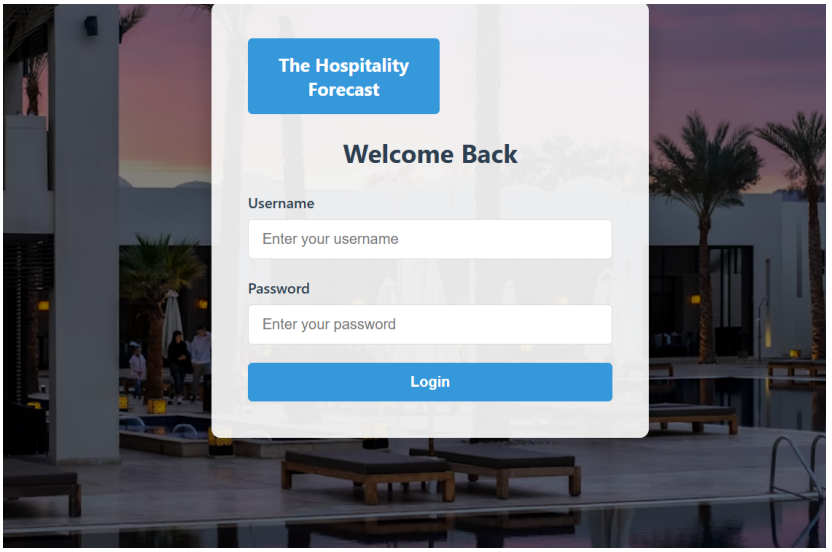

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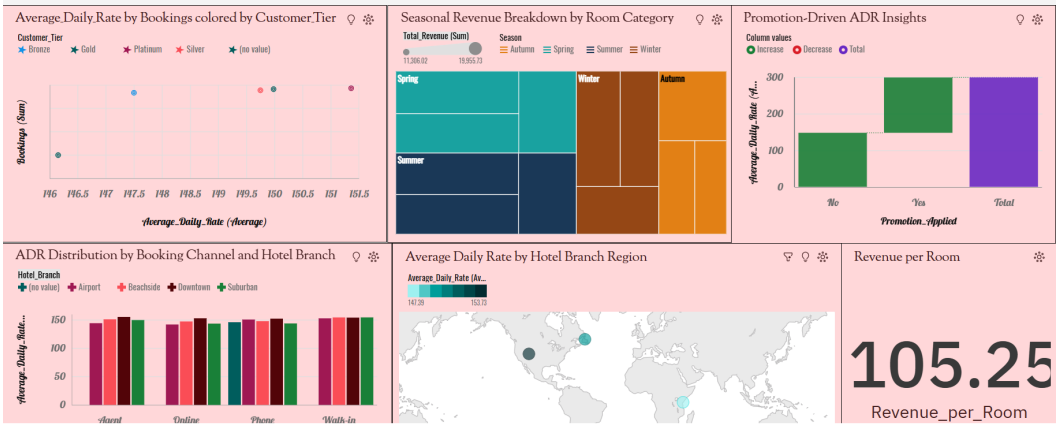
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